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In [1]: import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

from sklearn.model_selection import train_test_split
from sklearn.preprocessing import LabelEncoder
from sklearn.metrics import accuracy_score, precision_score, recall_score, f1_score

from sklearn.linear_model import LogisticRegression
from sklearn.ensemble import RandomForestClassifier
```

```
In [2]: df = pd.read_csv("train.csv")
df.head()
```

```
Out[2]:
```

	PassengerId	Survived	Pclass	Name	Sex	Age	SibSp	Parch	Ticket	Fare
0	1	0	3	Braund, Mr. Owen Harris	male	22.0	1	0	A/5 21171	7.2500
1	2	1	1	Cumings, Mrs. John Bradley (Florence Briggs Th...	female	38.0	1	0	PC 17599	71.2833
2	3	1	3	Heikkinen, Miss. Laina	female	26.0	0	0	STON/O2. 3101282	7.9250
3	4	1	1	Futrelle, Mrs. Jacques Heath (Lily May Peel)	female	35.0	1	0	113803	53.1000
4	5	0	3	Allen, Mr. William Henry	male	35.0	0	0	373450	8.0500

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In [4]: df['Age'] = df['Age'].fillna(df['Age'].median())
df['Embarked'] = df['Embarked'].fillna(df['Embarked'].mode()[0])
```

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In [5]: le = LabelEncoder()

df['Sex'] = le.fit_transform(df['Sex'])
df['Embarked'] = le.fit_transform(df['Embarked'])
```

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In [6]: X = df.drop("Survived", axis=1)
        y = df["Survived"]
```

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In [7]: X_train, X_test, y_train, y_test = train_test_split(
        X, y, test_size=0.2, random_state=42
        )
```

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In [8]: lr_model = LogisticRegression(max_iter=1000)
        lr_model.fit(X_train, y_train)

        y_pred_lr = lr_model.predict(X_test)
```

```
In [9]: print("Logistic Regression Performance:")
        print("Accuracy:", accuracy_score(y_test, y_pred_lr))
        print("Precision:", precision_score(y_test, y_pred_lr))
        print("Recall:", recall_score(y_test, y_pred_lr))
        print("F1 Score:", f1_score(y_test, y_pred_lr))
```

Logistic Regression Performance:
Accuracy: 0.8100558659217877
Precision: 0.7857142857142857
Recall: 0.7432432432432432
F1 Score: 0.7638888888888888

```
In [10]: rf_model = RandomForestClassifier(n_estimators=100, random_state=42)
        rf_model.fit(X_train, y_train)

        y_pred_rf = rf_model.predict(X_test)
```

```
In [11]: print("Random Forest Performance:")
        print("Accuracy:", accuracy_score(y_test, y_pred_rf))
        print("Precision:", precision_score(y_test, y_pred_rf))
        print("Recall:", recall_score(y_test, y_pred_rf))
        print("F1 Score:", f1_score(y_test, y_pred_rf))
```

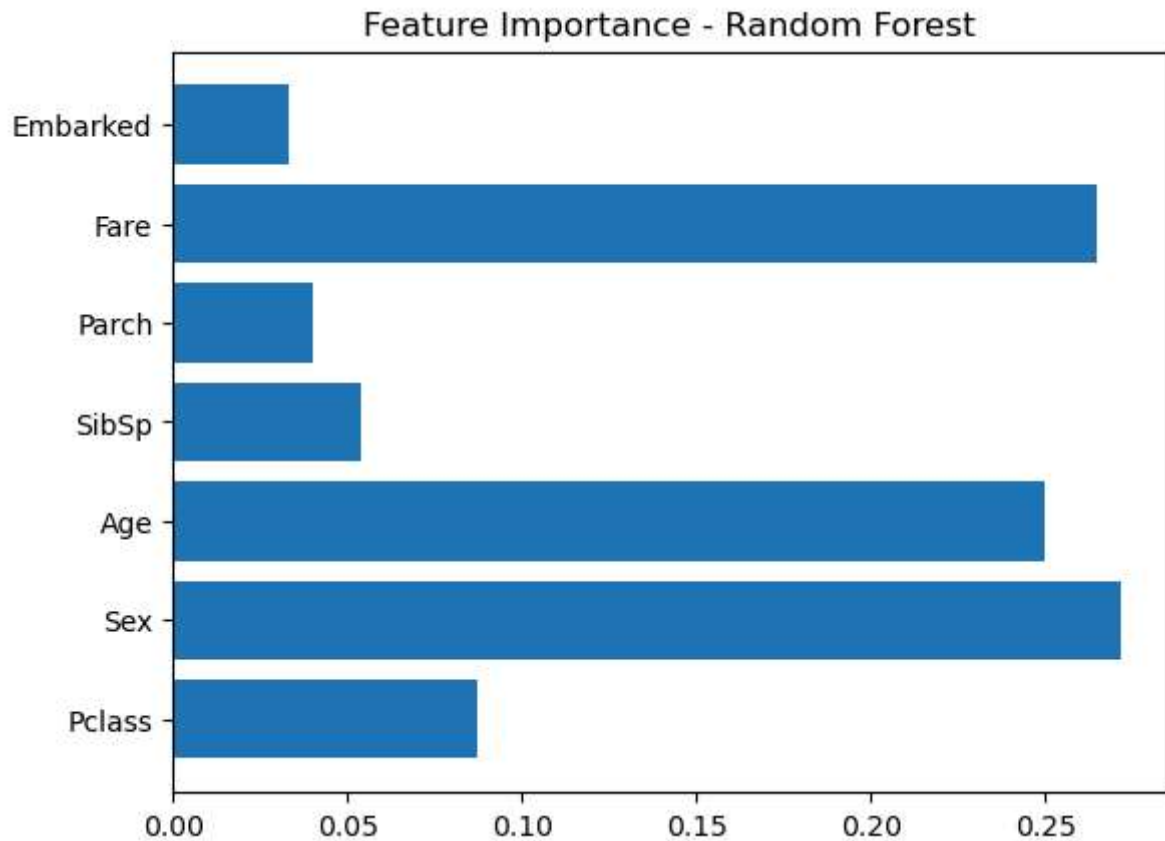
Random Forest Performance:
Accuracy: 0.8212290502793296
Precision: 0.8088235294117647
Recall: 0.7432432432432432
F1 Score: 0.7746478873239436

```
In [12]: print("Logistic Regression Confusion Matrix")
        print(confusion_matrix(y_test, y_pred_lr))

        print("Random Forest Confusion Matrix")
        print(confusion_matrix(y_test, y_pred_rf))
```

Logistic Regression Confusion Matrix
[[90 15]
 [19 55]]
Random Forest Confusion Matrix
[[92 13]
 [19 55]]

```
In [13]: importances = rf_model.feature_importances_  
features = X.columns  
  
plt.barh(features, importances)  
plt.title("Feature Importance - Random Forest")  
plt.show()
```



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In [ ]:
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